

2 =====

3 N=5, fc=868 MHz, BW=10 MHz, FR4 h=1.6 mm

4 =====

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8 PART A – HOW TO RUN stage2_final.py

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11 The script only needs Python 3 + numpy + matplotlib.

12 No other libraries required (skrf was removed).

14 OPTION 1 – You already have Python installed

15 -----

16 1. Open a terminal (Command Prompt on Windows, Terminal on Linux/Mac)

18 2. Check you have Python:

19 python --version (Windows)

20 python3 --version (Linux/Mac)

21 Should print Python 3.8 or newer.

23 3. Install the two required libraries if you haven't already:

24 pip install numpy matplotlib (Windows)

25 pip3 install numpy matplotlib (Linux/Mac)

26 Only need to do this once.

28 4. Run the script:

29 python stage2_final.py (Windows)

30 python3 stage2_final.py (Linux/Mac)

32 5. It will print results to the terminal and save these files

33 in a subfolder called "results/" next to the script:

34 stage2_response.png wideband S21/S11 + group delay

35 stage2_passband_zoom.png zoom into ±35 MHz of passband

36 stage2_vs_stage1.png circuit model vs ideal Chebyshev

37 stage2_network.s2p Touchstone file for QUCS-S import

38 stage2_dimensions.txt all physical dimensions

39 stage2_report.txt pass/fail summary

41 OPTION 2 – You don't have Python (easiest install)

42 -----

43 1. Download Python from <https://www.python.org/downloads/>

44 On Windows: tick "Add Python to PATH" during install.

46 2. Then follow steps 3-5 above.

48 OPTION 3 – Google Colab (no install at all)

49 -----

50 1. Go to <https://colab.research.google.com>

51 2. New notebook → click the + Code cell

52 3. Paste the entire stage2_final.py content into the cell

53 4. At the top, change this line:

54 OUT = Path("/home/claude/hairpin_bpf/results")

55 to:

56 OUT = Path("/content/results")

57 5. Click Run (Shift+Enter)

58 6. Download the files from the Files panel on the left

59

60 WHAT THE -44 dB IL MEANS

61 -----

62 The script will show S21 @ fc = -44 dB. This is NOT a bug.

63 It is the theoretical loss budget for FR4 at this narrow bandwidth.

64 The lossless result (0 dB) confirms the filter topology is correct.

65 See stage2_report.txt for the full explanation.

66 Real measured IL after QUCS-S EM tuning: expected 3-5 dB.

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PART B – QUCS-S 26.1 STEP-BY-STEP GUIDE

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73 CRITICAL FIRST STEP – SWITCH TO QucsatorRF

74 -----

75 Microstrip components (MLIN, MCOUPLED, MOPEN, MSUBST) only work

76 with the QucsatorRF backend. Ngspice does NOT support them in 26.1.

77

78 Menu: Simulation → Select default simulator

79 In the dropdown, choose: QucsatorRF

80 Click OK.

81

82 You will see "QucsatorRF" in the bottom status bar of the window.

83 If you don't see it, the microstrip components will silently fail.

84

85

86 STEP 1 – CREATE A NEW PROJECT

87 -----

88 File → New Project

89 Give it a name, e.g. "hairpin868"

90 Click OK – a new blank schematic opens.

91

92

93 STEP 2 – PLACE THE SUBSTRATE (do this first, on an empty area)

94 -----

95 Left panel: click the "Components" tab

96 In the dropdown at the top of that panel, select:

97 "Transmission Lines"

98

99 Scroll down until you see "Substrate" (labelled "SUBST").

100 Click it once → move mouse to schematic → left-click to place it

101 somewhere in the corner (it doesn't connect to anything).

102

Double-click the placed SUBST to open properties:

```
er    = 4.4
h     = 1.6 mm
t     = 0.035 mm
tand  = 0.02
rho   = 1.724e-8
D     = 1.5e-7      (surface roughness – leave default)
```

Name field at the top: type Subst1

Click OK.

STEP 3 – PLACE PORT 1 (left side)

Components dropdown → "Ports"

Click "Port" → place on schematic (left side, middle height)

Double-click → set:

```
Num = 1
Z    = 50 Ohm
```

Click OK.

STEP 4 – SHORT FEED LINE (port → first coupling gap)

Components dropdown → "Transmission Lines"

Click "MLIN" → place to the right of Port 1

Double-click → set:

```
Subst = Subst1
W      = 3.059 mm
L      = 5 mm
```

Click OK.

Draw a wire from Port 1 right terminal → left terminal of MLIN.

(Click the Wire button in the toolbar, or press W)

STEP 5 – INPUT COUPLING GAP (Port → Resonator 1)

Components dropdown → "Transmission Lines"

Click "MCOUPLED" → place to the right of the feed MLIN.

MCOUPLED has 4 terminals arranged like this:

```
1 ●—[upper line]—● 2
3 ●—[lower line]—● 4
```

The gap S is between the upper and lower lines.

Double-click → set:

```
Subst = Subst1
W      = 3.059 mm
L      = 47.348 mm  ← arm length (tune later)
```

S = 2.472 mm ← input coupling gap

Wiring:

Terminal 1 → connect to right end of feed MLIN

Terminal 2 → connect to GND (ground symbol from toolbar)

Terminal 3 → this is the start of Resonator 1, arm A (wire goes right)

Terminal 4 → leave open for now (or connect small open stub later)

STEP 6 – RESONATOR 1

Each resonator is a U-shape (two arms connected at the top).

6a. Arm A (top arm):

Components → Transmission Lines → MLIN

W = 3.059 mm, L = 47.348 mm, Subst = Subst1

Connect left end to Terminal 3 of the input MCOUPLED.

6b. Wire connecting the two arms at the top:

Draw a short wire from right end of Arm A going upward,

then left, forming the U-shape top.

(Or use MBEND components – optional for better accuracy.)

6c. Arm B (bottom arm, runs parallel to Arm A):

Components → Transmission Lines → MLIN

W = 3.059 mm, L = 47.348 mm, Subst = Subst1

Connect its left end (open end) to the wire from step 6b.

The RIGHT end of Arm B will connect to the next coupling gap.

6d. Open-end correction at the tip of Arm B:

Components → Transmission Lines → MOPEN

W = 3.059 mm, Subst = Subst1

Connect to the LEFT end of Arm B.

(MOPEN corrects the fringing capacitance at the open end,

which effectively shortens the resonant frequency slightly.)

NOTE: In QUCS-S, MOPEN has only ONE terminal – it connects

to one end of the line only. The other side is open.

STEP 7 – INTERIOR COUPLING GAPS (between resonators)

Between the right end of Arm B (Resonator 1) and Arm A (Resonator 2):

Components → Transmission Lines → MCOUPLED

W = 3.059 mm, L = 47.348 mm, S = 2.649 mm, Subst = Subst1

Terminal 1 → right end of Resonator 1, Arm B

Terminal 2 → open (or GND – this is the far end of the coupler)

Terminal 3 → start of Resonator 2, Arm A

Terminal 4 → open

205
206 Gap values for each interior coupler:
207 Between Res 1 and Res 2: $S = 2.649$ mm
208 Between Res 2 and Res 3: $S = 2.649$ mm
209 Between Res 3 and Res 4: $S = 2.649$ mm
210 Between Res 4 and Res 5: $S = 2.649$ mm
211
212
213 STEP 8 – RESONATORS 2 THROUGH 5
214 -----
215 Repeat Step 6 (a, b, c, d) for each of the 5 resonators.
216 All arm lengths start at 47.348 mm and all widths at 3.059 mm.
217
218 TIP: Copy-paste works well. Select Resonator 1 (Ctrl+A won't work –
219 drag a selection box around just the resonator parts), then Ctrl+C,
220 Ctrl+V, and place to the right. You'll need to re-wire the connections.
221
222
223 STEP 9 – OUTPUT COUPLING GAP AND PORT 2
224 -----
225 Mirror of Step 5:
226 MCOUPLED: W=3.059mm, L=47.348mm, S=2.472mm, Subst=Subst1
227 Terminal 1 → right end of Resonator 5, Arm B
228 Terminal 2 → open
229 Terminal 3 → short feed MLIN (W=3.059mm, L=5mm)
230 Terminal 4 → GND
231
232 Port 2 → Num=2, Z=50 Ohm → connect to right end of output feed MLIN.
233
234
235 STEP 10 – S-PARAMETER SIMULATION BLOCK
236 -----
237 Components dropdown → "Simulations"
238 Click "S-Parameter Simulation" (not AC – S-parameter is better here)
239 Place anywhere on the schematic (not connected to circuit).
240
241 Double-click → set:
242 Start frequency = 700 MHz (enter as: 700e6 or 700 MHz)
243 Stop frequency = 1050 MHz
244 Number of points = 701 (or more – 1401 for smoother curves)
245 Sweep type = linear
246
247 Click OK.
248
249
250 STEP 11 – RUN THE SIMULATION
251 -----
252 Press F2 (or menu: Simulation → Simulate)
253
254 A console window appears. Wait for it to finish.
255 If you see errors about "no substrate" → you forgot to set

Subst=Subst1 on one of the microstrip components.

If you see "unknown component" → you are on Ngspice, not QucsatorRF.

When done, click "Exit" to close the console.

STEP 12 – ADD DIAGRAMS TO VIEW RESULTS

Components dropdown → "Diagrams"

Click "Cartesian" → place on schematic.

Double-click the diagram → in the dialog:

Left panel shows available data variables.

Find and double-click: $\text{dB}(S[2,1])$ → adds S21 in dB

Also add: $\text{dB}(S[1,1])$ → adds S11 in dB

In the "Properties" tab:

Y-axis: min = -80, max = 5

X-axis: min = 700M, max = 1050M (in same units as your sweep)

Click OK.

Add a second Cartesian diagram zoomed in:

X: 855 MHz to 885 MHz

Y: -80 to 5 dB

This shows the passband detail.

STEP 13 – PLACE MARKERS

Click the Marker button in the toolbar (looks like a small flag)

Click on the S21 trace at 868 MHz.

The marker box shows: frequency, S21 value in dB.

Target: S21 > -6 dB (realistic for FR4)

S11 < -15 dB in passband

PART C – TUNING IN QUCS-S 26.1

USE THE TUNING SLIDER (fastest method)

Menu: Simulation → Tuning (or press F5)

A "Tuning" panel appears. Click on any parameter value on the schematic (e.g. the $L=47.348\text{mm}$ of an MLIN) – it appears in the panel with a slider.

Drag the slider → QUCS-S re-simulates automatically → plot updates.

TUNE IN THIS ORDER:

1. Centre frequency (most important)

The passband will be shifted LOW by ~5-15 MHz due to open-end fringing adding electrical length.

→ Select the L parameter of any resonator arm MLIN

→ Decrease L from 47.348 mm in 0.5 mm steps

→ Watch the S21 peak move right toward 868 MHz

→ Apply the SAME change to ALL 10 arms (5 resonators × 2 arms)

Typical result: $L \approx 44-46$ mm after correction.

2. Return loss (S11)

If S11 in passband is worse than -15 dB:

→ Tune S of the input/output MCOUPLED (S01 = 2.472 mm starting point)

→ Decrease S → stronger coupling → better return loss

→ Aim for $S_{11} < -20$ dB across 863-873 MHz

3. Bandwidth

If 3 dB BW is too wide: increase interior gap S values

If 3 dB BW is too narrow: decrease interior gap S values

→ Keep $S_{12}=S_{45}$, $S_{23}=S_{34}=S_{34}$ (symmetric filter)

→ Adjust in 0.05 mm steps

→ Target: 3 dB BW = 10 MHz

4. Ripple equalisation

If passband looks tilted (deeper on one side):

→ Slightly break symmetry on S_{12} vs S_{45} by ± 0.1 mm

→ Usually not needed if layout is symmetric

PART D – IMPORT stage2_network.s2p FOR COMPARISON

To overlay the Python Stage 2 result with your QUCS-S result:

1. Menu: File → Load Data File

Navigate to stage2_network.s2p, click Open.

It appears in the "Content" tab on the left under "Data".

2. In your Cartesian diagram, double-click to edit it.

In the left variable list you will now see two datasets:

- Your simulation (e.g. $SP1.dB(S[2,1])$)

- The .s2p data (e.g. $stage2_network.dB(S[2,1])$)

Add both to the same diagram.

3. You will see the Python circuit model vs your QUCS-S physical model.

The QUCS-S curve should show higher S21 (less loss) because it properly accounts for the gap coupling, not just stub loss.

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PART E – EXPORT FINAL VALUES FOR PCB

After tuning is done, record the final values:

Arm length (all equal after tuning): _____ mm

Fold gap (inside each U): _____ mm

S01 input/output gap: _____ mm

S12, S45 gap: _____ mm

S23, S34 gap: _____ mm

Line width W (fixed): 3.059 mm

These are the numbers you put into KiCad / EasyEDA for PCB layout.

To export S-parameters from QUCS-S:

Menu: File → Export Data → Touchstone (*.s2p)

This is your Stage 3 verified result.

PART F – EXPECTED PERFORMANCE ON FR4 (realistic targets)

Parameter	Expected after QUCS-S tuning (FR4)
Insertion loss (IL)	3 – 5 dB (FR4 $\tan \delta = 0.02$ limits this)
Return loss	> 15 dB in passband
3 dB bandwidth	~ 10 MHz (tune to spec)
Stopband ±50 MHz	> 30 dB rejection
Board area	~ 50 × 30 mm (5 hairpins at 868 MHz on FR4)

The IL is dominated by FR4 loss and CANNOT be improved by tuning.

It is a fundamental material constraint at FBW=1.15%, N=5.

If your application cannot absorb 3-5 dB IL, the only fixes are:

- Add an LNA before the filter to compensate
- Use Rogers 4003C substrate → expect ~0.8 dB IL instead

END OF GUIDE